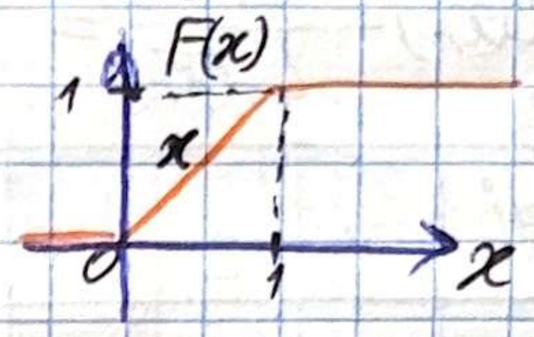
№11

а) Построить оптим. крит. по выборке объёма $n$

с крит. обл. $G$: $x_{min} < c$ и ур. знач. $\alpha$. $\alpha_1$ - ? $\alpha_2$ - ? $W$ - ? сост. крит. ?

$\alpha_2 \to \min$
$\alpha_1 \le \alpha$

$$\alpha_1 = P(x_{min} \le c \mid H_0) = 1-(1-F(c))^n \le \alpha$$




$$1-(1-c)^n \le \alpha$$

$$\Rightarrow 1-\sqrt[n]{1-\alpha} \ge c$$

$$\alpha_2 = P(x_{min} > c \mid H_1) = 1-P(x_{min} \le c \mid H_1) = 1-(1-(1-\Phi(c)))^n =$$

$$= (1-\Phi(c))^n = \left\langle\langle \Phi(c) = \int_0^c \frac{e^{1-x}}{e-1}\,dx = \frac{e}{e-1}(1-e^{-c}) \right\rangle\rangle = \left(1-\frac{e}{e-1}(1-e^{-c})\right)^n$$

$\alpha_2 \to \min$
$\alpha_1 \le \alpha$

$$\left(1-\frac{e}{e-1}(1-e^{-c})\right)^n \to \min \qquad \frac{d\alpha_2}{dc} = n\left(1-\frac{e}{e-1}(1-e^{-c})\right)^{n-1}\cdot\left(-\frac{e^{1-c}}{e-1}\right) = 0$$

$$1-\sqrt[n]{1-\alpha} \ge c \qquad \frac{e}{e-1}(1-e^{-c}) = 1$$

$$e^{-c} = e^{-1} \Rightarrow c' = 1$$

т.к. $1-\alpha \in (0,1)$, то $\sqrt[n]{1-\alpha} \in (0,1)$

и $1-\sqrt[n]{1-\alpha} \in (0,1) \Rightarrow c' > 1-\sqrt[n]{1-\alpha}$

$$\frac{d^2\alpha_2}{dc^2} = n(n-1)\left(1-\frac{e}{e-1}(1-e^{-c})\right)^{n-2}\left(\frac{-e^{1-c}}{e-1}\right)^2 + n\left(1-\frac{e}{e-1}(1-e^{-c})\right)^{n-1}\left(\frac{e^{1-c}}{e-1}\right) > 0$$

$\Rightarrow c'$ - т. min $\Rightarrow$ т.к. $c \le 1-\sqrt[n]{1-\alpha}$, то

$$\Rightarrow c = 1-\sqrt[n]{1-\alpha} \Rightarrow G: x_{min} < 1-\sqrt[n]{1-\alpha}$$
$(\alpha_1 = \alpha)$

Мощность $W = P(x_{min} < 1-\sqrt[n]{1-\alpha} \mid H_1) = 1-(1-\Phi(c))^n =$

$$= 1-\left(1-\frac{e}{e-1}(1-e^{-c})\right)^n = 1-\left(1-\frac{e}{e-1}\left(1-e^{-1+\sqrt[n]{1-\alpha}}\right)\right)^n \ominus$$

$$\left\langle\langle -1+\sqrt[n]{1-\alpha} = -1+e^{\frac{1}{n}\ln(1-\alpha)} = -1+1+\frac{1}{n}\ln(1-\alpha)+o\left(\frac{1}{n}\right) = \frac{1}{n}\ln(1-\alpha)+o\left(\frac{1}{n}\right)\right\rangle\rangle$$

$$e^{-1+\sqrt[n]{1-\alpha}} = e^{\frac{1}{n}\ln(1-\alpha)+o\left(\frac{1}{n}\right)} = 1+\frac{1}{n}\ln(1-\alpha)+o\left(\frac{1}{n}\right)$$

$$\ominus\ 1-\left(1-\frac{e}{e-1}\left(1-1-\frac{1}{n}\ln(1-\alpha)+o\left(\frac{1}{n}\right)\right)\right)^n = 1-\left(1+\frac{e}{e-1}\frac{1}{n}\ln(1-\alpha)+o\left(\frac{1}{n}\right)\right)^n \to$$

$$\xrightarrow{n\to\infty} 1-e^{\frac{e\ln(1-\alpha)}{e-1}} = 1-(1-\alpha)^{\frac{e}{e-1}} \ne 1 \Rightarrow \text{крит. не сост.}$$

№ T12 $H_0: p_4 = \frac{8}{24}, \; p_3 = \frac{4}{24}, \; p_1 = p_2 = \frac{6}{24}$ $\alpha = 0{,}2$

$H_1: p_1 = p_2 = p_3 = p_4 = \frac{6}{24}$ $n = 2$

Найти наиб. мощный крит., найти мощность крит.

$$l(\vec{x_n}) = \frac{L_1(\vec{x_n})}{L_0(\vec{x_n})}$$

$p_0$:

| $H_0$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1/6 | 1/6 | 1/24 | 1/12 |
| 2 | 1/6 | 1/6 | 1/24 | 1/12 |
| 3 | 1/24 | 1/24 | 1/36 | 1/18 |
| 4 | 1/12 | 1/12 | 1/18 | 1/9 |

$p_1$:

| $H_1$: | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1/16 | 1/16 | 1/16 | 1/16 |
| 2 | 1/16 | 1/16 | 1/16 | 1/16 |
| 3 | 1/16 | 1/16 | 1/16 | 1/16 |
| 4 | 1/16 | 1/16 | 1/16 | 1/16 |

$l$:

| $l$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1 | 1 | 3/2 | 3/4 |
| 2 | 1 | 1 | 3/2 | 3/4 |
| 3 | 3/2 | 3/2 | 9/4 | 9/8 |
| 4 | 3/4 | 3/4 | 9/8 | 9/16 |

$p_{ij} = p_i \cdot p_j$ $P(l \geq C \mid H_0) \leq 0{,}2$

берём max $\left(\frac{9}{4}\right)$, смотрим $p_0 = \frac{1}{36}$, это мало ($\leq 2$), идём дальше

и так далее... $\frac{1}{36} + \frac{1}{24} + \frac{1}{24} + \frac{1}{24} + \frac{1}{24} = \frac{7}{36} < 0{,}2$

след ($\frac{9}{8}$) при добавлении будет перебор ($> 0{,}2$)

$\Rightarrow G: l \geq C$, попадают все элем. $\geq \frac{3}{2} \Rightarrow C = \frac{3}{2}$

$G: l \geq \frac{3}{2}$ — крит. обл.

$\Rightarrow w = 5 \cdot \frac{1}{16} = \frac{5}{16}$ $\alpha_1 = \frac{7}{36}$ $\alpha_2 = 1 - w = \frac{11}{16}$

<u>Ответ:</u> $w = \frac{5}{16}$

№ Т13

$\alpha = 0{,}05$

Египет:
$n = 139$ черепов $\quad \tilde{\sigma}_{дл.} = 5{,}722$ мм $\quad \tilde{\sigma}_{шир.} = 4{,}612$ мм

Европа:
$m = 1000$ $\quad \tilde{\sigma}_{дл.} = 6{,}161$ мм $\quad \tilde{\sigma}_{шир.} = 5{,}055$ мм

$W - ?$

$(дл)\xi \sim N(\theta_1, \theta_2^2)$, $\quad (шир)\eta \sim N(\theta_1', \theta_2'^2)$

1) $H_0: \theta_{2\,евр.}^2 = \theta_{2\,егип.}^2$ $\quad$ 2) $H_0: \theta_{2\,евр.}'^2 = \theta_{2\,егип.}'^2$

длина $\quad$ ширина

$H_1: \theta_{2\,евр.}^2 \neq \theta_{2\,егип.}^2$ $\quad H_1: \theta_{2\,евр.}'^2 \neq \theta_{2\,егип.}'^2$

$S^2 = \frac{n}{n-1}\tilde{\sigma}^2$ $\quad S^2_{дл.\,ег.} = 32{,}9785$ $\quad S^2_{шир.\,ег.} = 21{,}4246$

$S^2_{дл.\,евр.} = 37{,}9959$ $\quad S^2_{шир.\,евр.} = 25{,}5786$

$\tilde{\Delta}_{дл.} = \frac{S^2_{дл.\,ег.}}{S^2_{дл.\,евр.}} = 0{,}867795$ $\quad \tilde{\Delta}_{шир.} = \frac{S^2_{шир.\,ег.}}{S^2_{шир.\,евр.}} = 0{,}8376$

$\tilde{\Delta} \sim F(n-1, m-1) = F(138, 999)$

$\Rightarrow G: \tilde{\Delta} \le F_{\frac{\alpha}{2}}(138, 999) \underset{или}{\cup} \tilde{\Delta} \ge F_{1-\frac{\alpha}{2}}(138, 999)$

$F_{\frac{\alpha}{2}}(138, 999) = 0{,}76738$ $\quad F_{1-\frac{\alpha}{2}}(138, 999) = 1{,}27169$

$\Rightarrow$ обе $\tilde{\Delta}$ (длины и шир.) $\notin G \Rightarrow$ нет оснований отв. $H_0$ в обеих случаях

$W = P(\Delta \in G | H_1) = P(\tilde{\Delta} \le F_{\frac{\alpha}{2}}(138, 999) | H_1) + P(\tilde{\Delta} \ge F_{1-\frac{\alpha}{2}}(138, 999) | H_1) =$

$= P\left(\frac{S^2_{ег.}}{S^2_{евр.}} \cdot \frac{\tilde{\sigma}^2_{евр.}}{\underbrace{\tilde{\sigma}^2_{ег.}}_{\theta}} \le F_{\frac{\alpha}{2}}(138, 999)\right) + P\left(\frac{S^2_{ег.}}{S^2_{евр.}} \cdot \frac{\tilde{\sigma}^2_{евр.}}{\underbrace{\tilde{\sigma}^2_{ег.}}_{\theta}} \ge F_{1-\frac{\alpha}{2}}(138, 999)\right) \Rightarrow$

$\Rightarrow W = \int_{\frac{F_{\frac{\alpha}{2}}(138,999)}{\theta}}^{+\infty} q(t)\,dt + \int_{0}^{\frac{F_{1-\frac{\alpha}{2}}(138,999)}{\theta}} q(t)\,dt$ $\quad$ (см. Т13. рграфик. $W(\theta)$)

$\boxed{\sim T19}$ $\xi \sim N(a, \sigma_x^2)$ $\sigma_x^2 = 2$ $\vec{x}_n = [-1.11, -6.10, 2.42]$

$\sigma_y^2 = 1$ $\vec{y}_m = [-2.29, -2.91]$

$\eta \sim N(b, \sigma_y^2)$

$n = 3$ $m = 2$ $\alpha = 0.05$

$H_0: a = b$ по (T) Фишера $\frac{\bar{x} - a}{\sigma_x}\sqrt{n} \sim N(0,1)$

$H_1: a > b$

$\frac{\bar{y} - b}{\sigma_y}\sqrt{m} \sim N(0,1)$

$$\left.\begin{aligned} \bar{x} - a &\sim N\left(0, \frac{\sigma_x^2}{n}\right) \\ \bar{y} - b &\sim N\left(0, \frac{\sigma_y^2}{m}\right) \end{aligned}\right\} \Rightarrow \bar{x} - \bar{y} - (a - b) \sim N\left(0, \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}\right)$$

$$\Rightarrow \Delta = \frac{\bar{x} - \bar{y} - (a - b)}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} \sim N(0,1) \qquad G: \Delta \geq U_{1-\alpha}$$

$$\Delta \overset{H_0}{=} \frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} \sim N(0,1)$$

Пусть $a - b = \theta$

$$\Rightarrow \bar{x} - \bar{y} \overset{H_1}{\sim} N\left(\theta, \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}\right)$$

$$\Rightarrow W = P(\Delta \in G \mid H_1) = P\left(\Delta \geq U_{1-\alpha}\left(0, \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}\right)\right)$$

(график для $W(\theta)$ в T14, ipynb)